Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	06-04-2026
Team ID	69bd2f9cc71a9ebd6a50e3b1
Project Name	Smart library request workflow in service now
Maximum Marks	4 Marks

Technical Architecture:

The Smart Library Request Workflow is designed using a **3-tier architecture** in ServiceNow:

- **Presentation Layer:** ServiceNow Portal / Web UI for students and librarians
- **Application Layer:** ServiceNow workflows, business rules, and automation engine
- **Data Layer:** ServiceNow database (tables for users, books, requests, transactions)

Guidelines:

1. User logs into ServiceNow portal
2. Searches for books and checks availability
3. Raises a book request
4. Request goes to librarian for approval
5. Approval/Rejection is processed
6. Notification sent to user
7. Book issue/return is recorded in system

External Interfaces:

- Email Notification Service
- SMS Gateway API (for alerts)
- Optional: Barcode/QR Scanner integration

Data Storage:

- ServiceNow Tables (User, Book, Request, Transaction)
- Cloud-based storage (ServiceNow platform)

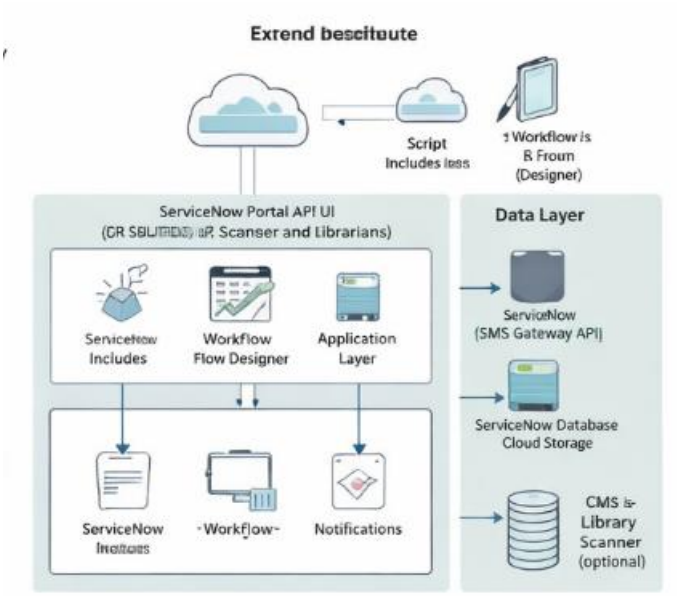

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for students & librarians	ServiceNow Portal, HTML, CSS, JavaScript
2.	Application Logic-1	Book search and availability logic	ServiceNow Script Includes, Glide API
3.	Application Logic-2	Request workflow and approval process	ServiceNow Flow Designer / Workflow
4.	Application Logic-3	Notification and alert system	ServiceNow Notifications, Email API
5.	Database	Stores users, books, requests data	ServiceNow Tables (Glide DB)
6.	Cloud Database	Cloud-based data management	ServiceNow Cloud Platform
7.	File Storage	Store book details, reports	ServiceNow Attachment System
8.	External API-1	Email notification service	SMTP / ServiceNow Email Integration
9.	External API-2	SMS notification service	Twilio API / SMS Gateway
10.	Machine Learning Model	Book recommendation (optional)	ServiceNow Predictive Intelligence
11.	Infrastructure (Server / Cloud)	Deployment environment	ServiceNow SaaS Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Uses web technologies integrated with ServiceNow	HTML, CSS, JavaScript
2.	Security Implementations	Role-based access, authentication, encryption	ServiceNow ACL, OAuth, SSL Encryption
3.	Scalable Architecture	3-tier scalable architecture	ServiceNow Cloud, Microservices
4.	Availability	High availability with cloud support	ServiceNow Cloud (99.9% uptime)
5.	Performance	Fast response using optimized workflows	Caching, Indexed DB, Efficient Queries

References:

- https://c4model.com/
- https://developer.ibm.com/patterns/
- https://www.ibm.com/cloud/architecture/
- https://aws.amazon.com/architecture
- https://medium.com/